Zhijie Yan

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Education

Ph.D. in Computer Science, The Chinese University of Hong Kong, Shenzhen — Shenzhen, China
Supervisor: ***Prof. Li Jiang*** — *Sept 2026 – Present*

- **Research Interests:** World Models, 4D Scene Understanding, Autonomous Driving, and Robotics

Master in Mechanical Engineering, Beihang University — Beijing, China
Supervisor: ***Prof. Zuoxu Wang*** — *Sept 2023 – Jun 2026*

- Rank: **1 / 186**; Thesis: *Mobile Robot Navigation and Manipulation Based on Dynamic 3D Scene Representation*

Software Engineering, Jiangxi University of Finance and Economics — Jiangxi, China
Supervisor: ***Dr. Jifu Chen*** — *Sept 2019 – Jun 2023*

- GPA: **93.48 / 100**, Rank: **5 / 436**
- Thesis: *Trajectory Prediction via Global Intention Localization and Local Motion Refinement*

Research Interests

My research centers on **World Models** and **4D Scene Understanding** for modeling and predicting dynamic physical environments, with applications to **Autonomous Driving** and **Robotics**. I am particularly interested in building spatial-temporal representations and generative models that enable intelligent agents to perceive, predict, plan, and interact effectively in complex real-world environments.

Publications († denotes corresponding author.)

Journal Articles

Dynamic Open-Vocabulary 3D Scene Graphs for Long-term Language-Guided Mobile Manipulation
Zhijie Yan, Shufei Li, Zuoxu Wang†, Lixiu Wu, Han Wang, Jun Zhu, Lijiang Chen, Jihong Liu
IEEE Robotics and Automation Letters ***(RA-L, JCR-Q1, IF: 5.3), 2025***

DREAM: Dynamic Resilient Spatio-Semantic Memory with Hybrid Localization for Mobile Manipulation
Zhijie Yan, Shufei Li, Ze Zhang, Xin Liu, Yuhang Zheng, Zuoxu Wang†
IEEE/ASME Transactions on Mechatronics ***(TMECH, JCR-Q1, IF: 6.3), 2026****; IROS Workshop* ***(Best Paper Award), 2025***

IndVisSGG: VLM-based scene graph generation for industrial spatial intelligence
Zuoxu Wang, **Zhijie Yan**, Shufei Li†, Jihong Liu
Advanced Engineering Informatics ***(ADVEI, JCR-Q1, IF: 9.9), 2025***

VLM-MSGraph: Vision Language Model-enabled Multi-hierarchical Scene Graph for robotic assembly
Shufei Li, **Zhijie Yan**, Zuoxu Wang†, Yiping Gao
Robotics and Computer-Integrated Manufacturing ***(RCIM, JCR-Q1, IF: 11.4), 2025***

Conference Papers

INT2: Interactive Trajectory Prediction at Intersections
Zhijie Yan, Pengfei Li, Zheng Fu, Shaocong Xu, Yongliang Shi, Xiaoxue Chen, Yuhang Zheng, Yang Li, Tianyu Liu, Chuxuan Li, Nairui Luo, Xu Gao, Yilun Chen, Zuoxu Wang, Yifeng Shi, Pengfei Huang, Zhengxiao Han, Jirui Yuan, Jiangtao Gong, Guyue Zhou, Hang Zhao, Hao Zhao†
International Conference on Computer Vision ***(ICCV), 2023***

ManufVisSGG: A Vision-Language-Model Approach for Cognitive Scene Graph Generation in Manufacturing Systems
Zhijie Yan, Zuoxu Wang†, Shufei Li, Mingrui Li, Xinxin Liang, Jihong Liu
International Conference on Automation Science and Engineering ***(CASE), 2024***

Long-term Interactive Driving Simulation: MPC to the Rescue
Zhengxiao Han, **Zhijie Yan**, Yang Li, Pengfei Li, Yifeng Shi, Nairui Luo, Xu Gao, Yongliang Shi, Pengfei Huang, Jiangtao Gong, Guyue Zhou, Yilun Chen, Hang Zhao, Hao Zhao†
CAAI International Conference on Artificial Intelligence ***(CICAI), 2023***

Large Language Models Powered Context-aware Motion Prediction
Xiaoji Zheng, Lixiu Wu, **Zhijie Yan**, Yuanrong Tang, Hao Zhao, Chen Zhong, Bokui Chen, Jiangtao Gong†
*International Conference on Intelligent Robots and Systems **(IROS), 2024***

TOD3Cap: Towards 3D Dense Captioning in Outdoor Scenes
Bu Jin, Yupeng Zheng†, Pengfei Li, Weize Li, Yuhang Zheng, Sujie Hu, Xinyu Liu, Jinwei Zhu, **Zhijie Yan**, Haiyang Sun, Kun Zhan, Peng Jia, Xiaoxiao Long, Yilun Chen, Hao Zhao
*European Conference on Computer Vision **(ECCV), 2024***

Internship Experience

Nov 2026 – Present: Research Intern, Yinwang, **Huawei**, Shenzhen, China

Feb 2025 – July 2025: Research Intern, Reinforcement Learning Group, **NoetixRobotics**, Beijing, China

Jul 2022 – Nov 2023: Research Intern, DISCOVER LAB, AIR, **Tsinghua University**, Beijing, China

Awards

Outstanding Graduate of Beihang University
School-level honor, 2026

National Scholarship
National-level scholarship, 2025

Best Paper Award at IROS 2025 Workshop
International-level scholarship, 2025

First Prize Scholarship for Graduate Students
School-level scholarship, 2025

China College Students' Self-Reliance Star
National-level scholarship, 2022

The Second Prize of the 23rd China Robot and Artificial Intelligence Competition
Ministry of Education national-level competition award, 2021

The Second Prize of the 14th China College Students' Computer Design Competition
Ministry of Education national-level competition award, 2021

Skills

PROGRAMMING : Python / C++ / Java, Web (JavaScript, HTML/CSS), ROS/ROS2

TOOLS : Ubuntu, SolidWorks, Isaac Sim, MuJoCo, Isaac Gym, Genesis

LANGUAGES : English (Fluent) / Mandarin (Native)